

**BECM 3115**

# **CLIMATE & ARCHITECTURAL DESIGN**

## **FORMULA • THEORY • PROOF / DERIVATIONS**

Exam-focused master booklet | বাংলা ব্যাখ্যা | Source-safe formulas | Repeated Question Bank mapping

### **What this booklet contains**

1) Master Formula Sheet 2) Section A + Section B high-frequency theory capsules 3) Step-by-step proof/derivations 4) Repeated question list with years 5) Last-minute revision pages.

Primary basis: uploaded lecture slides, class notes, regular final-term question papers (2016-2024), and uploaded reference books only where needed for cross-checking.

**Made by 2223033**

## কীভাবে এই বই ব্যবহার করবে | Accuracy & Exam-use Guide

### Formula status legend

GREEN / SOURCE = uploaded slide/note-এ formula সরাসরি আছে। GOLD / DERIVED = uploaded definition বা repeated numerical থেকে পরীক্ষায় ব্যবহারের জন্য relation step-by-step derive করা হয়েছে। RED / QB-ONLY = question bank-এ numerical/term আছে, কিন্তু uploaded slide-এ formula নেই; সেখানে formula invent করা হয়নি।

## Exam strategy

1. Formula প্রথমে symbol + unit সহ শিখবে; তারপর ১টি derivation বুঝবে।
2. Theory answer: Definition mechanism /factors table /sketch design significance conclusion .
3. Proof/derivation: Given concept assumptions /definition algebra /energy balance final boxed relation.
4. প্রতি Section-এ ৪ set থেকে ৩ set answer করতে হয়; তাই repeated theory + derivation/numerical mix প্রস্তুত রাখবে.

## Important source caution

### No invented shading formula

2023-2024 papers-এ solar shading projection numerical আছে; uploaded Shading Device slide-এ horizontal/vertical/combined types আছে, কিন্তু projection calculation formula নেই। তাই এই booklet numerical formula বানিয়ে দেয়নি। Instructor-verified sun-angle method থাকলে সেটাই follow করবে.

## Contents

| Part | Coverage                                      |
|------|-----------------------------------------------|
| I    | Master Formula Sheet                          |
| II   | Current Section A - Shahriar Sir topics       |
| III  | Current Section B - Billah/Safayet Sir topics |
| IV   | Proof & Derivation Bank                       |
| V    | Most Repeated Theory Question Bank            |
| VI   | Last-minute 2-page revision                   |

# PART I — MASTER FORMULA SHEET

| Formula / Relation                                                 | Meaning / Unit                                                                              | Status  |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------|
| $RH = (AH / SH) \times 100\%$                                      | Relative humidity; AH & SH same basis (g/kg or g/m <sup>3</sup> )                           | SOURCE  |
| $P = P_a + P_v$                                                    | Atmospheric pressure = dry-air partial pressure + vapour partial pressure; N/m <sup>2</sup> | SOURCE  |
| $I\beta = I_n \cos \beta$                                          | Cosine law: intensity on tilted surface                                                     | SOURCE  |
| $DRI = R \times V_w$                                               | Driving Rain Index; R in m, V <sub>w</sub> in m/s → m <sup>2</sup> /s                       | SOURCE  |
| $Met - E_{vp} \pm C_{nd} \pm C_{nv} \pm Rad = 0$                   | Human thermal steady-state heat balance                                                     | SOURCE  |
| $Air\text{-}space/person = (V_{space} - V_{people/furniture}) / N$ | Available air volume per user; m <sup>3</sup> /person                                       | SOURCE  |
| $Q_{total} = N \times q_{person}$                                  | Total fresh-air requirement; L/s or m <sup>3</sup> /s                                       | DERIVED |
| $ACH = 3600 Q / V$                                                 | Air changes per hour, Q in m <sup>3</sup> /s, V in m <sup>3</sup>                           | DERIVED |
| $\dot{Q}_{air} = \rho V n c_p \Delta T / 3600$                     | Heat loss by n air changes/hour; W if SI units used                                         | DERIVED |

## Shortcut - Three anchors

Humidity: AH/SH. Radiation: cos. Ventilation:  $Q \times 3600 / V$ . এই তিনটা numerical family সবচেয়ে useful.

## Key constants / course values to remember

| Item                        | Course value / range                    | Why useful                     |
|-----------------------------|-----------------------------------------|--------------------------------|
| Solar constant              | 1395 W/m <sup>2</sup> (slide value)     | Global solar radiation theory  |
| Earth axis to orbital plane | 66.5° (equiv. 23.5° from perpendicular) | Seasons / Tropics              |
| Human deep-body temperature | around 37°C                             | Thermal comfort / heat balance |
| Warm-humid RH               | about 75%, may vary 55-100%             | Climate comparison             |
| Warm-humid annual rain      | about 2000-5000 mm                      | Climate comparison             |
| Hot-dry RH                  | about 10-55%                            | Climate comparison             |
| Hot-dry diurnal range       | about 17-22°C                           | Climate comparison             |
| DRI zones                   | s 3 heltered ; 3-7 moderate; >7 severe  | Driving-rain design            |

## PART II — CURRENT SECTION A | Shahriar Sir Topics

### Section A map

Climate-Context-Comfort • Built environment • Human thermal comfort • Comfort indices • Regional/site climate • Design specification • Microclimate • Shading • Climate-friendly movements • Ventilation / air-space / infiltration.

## A1. Climatic Comfort Indices

### Effective Temperature (ET)

- Different air-temperature + humidity combinations can feel equally comfortable.
- ET scale combines Temperature + Humidity + Air movement on a psychrometric basis.
- Radiation effect is not included in basic ET.

### Corrected Effective Temperature (CET)

- ET scale is corrected by incorporating radiation effect.
- Hence CET is more complete than ET where radiant environment matters.

### Equivalent Warmth (EW) & Operative Temperature (OT)

- EW correlates air temperature, humidity, mean radiant temperature with skin/clothing surface temperature.
- OT combines effects of radiation and air temperature; humidity and air movement are treated as negligible in the lecture slide.

#### Shortcut - ET-CET

$ET = T + H + \text{Air movement}$ .  $CET = ET + \text{Radiation}$ .

#### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                                    |
|------|---------------------------------------------------------------------|
| 2016 | ET and CET / bioclimatic chart; factors of climate comfort.         |
| 2017 | Corrected Effective Temperature; atmospheric depletion; cosine law. |
| 2021 | Comfort instruments/indices: ET, CET, EW; humidity definitions.     |
| 2023 | Origin of ET and how comfort indices were developed.                |

## A2. Relative Humidity

$$RH = (AH / SH) \times 100 \%$$

AH = actual/absolute humidity; SH = saturation humidity at same temperature.

*Directly stated in uploaded source*

বাংলায়: বাতাস যতটুকু জলীয়বাষ্প বাস্তুবে ধারণ করেছে (AH), সেই তাপমাত্রায় সর্বোচ্চ যতটুকু ধারণ করতে পারত (SH) - তার শতকরা অনুপাতই RH।

#### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                                             |
|------|------------------------------------------------------------------------------|
| 2020 | Given AH and saturation humidity, determine RH and required moisture change. |
| 2021 | Define AH, SH, RH and solve relation-based numerical.                        |

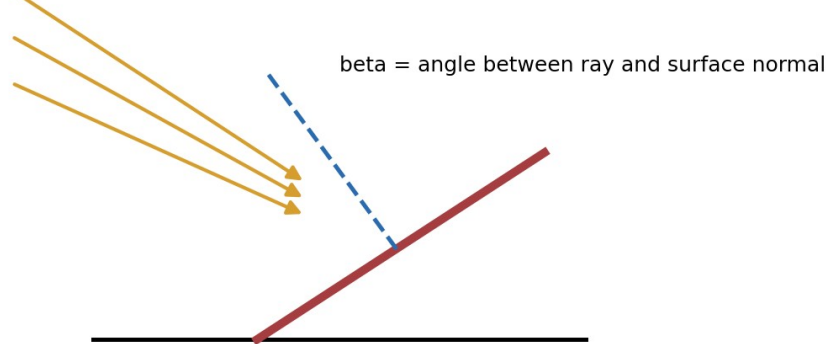
### A3. Cosine Law of Solar Radiation

$$I_{\beta} = I_n \times \cos \beta$$

$\beta$  = angle of incidence measured from surface normal.

*Directly stated in uploaded source*

**Cosine Law - oblique incidence spreads the same energy over a larger area**



$$I_{\text{tilted}} = I_{\text{normal}} \times \cos(\beta)$$

Oblique incidence spreads the same radiation over a larger area.

#### Shortcut - Normal = Maximum

$\beta = 0^\circ$  হলে  $\cos\beta = 1$  intensity maximum.  $\beta$  বাড়লে  $\cos\beta$  কমে  $\rightarrow$  received intensity কমে।

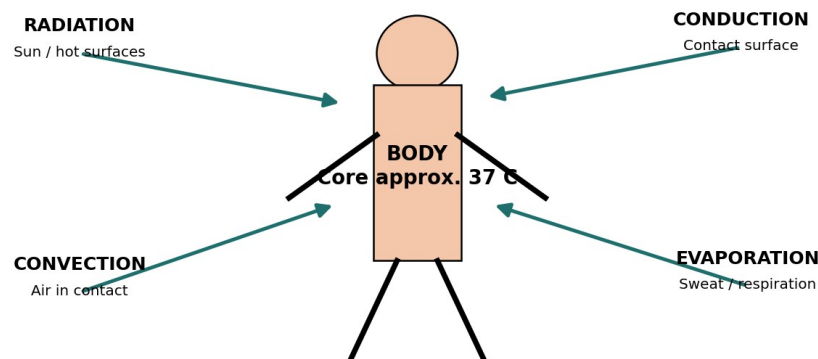
#### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                                        |
|------|-------------------------------------------------------------------------|
| 2017 | Write short note on Cosine Law.                                         |
| 2022 | Illustrate cosine law and explain use in designing buildings.           |
| 2024 | Describe cosine law with a diagram and show its use in building design. |

### A4. Human Heat Exchange & Heat Loss

**Human heat exchange: heat can be gained or lost depending on temperature gradients**



Conduction, convection, radiation and evaporation are the main exchange modes.

- Conduction: direct contact surface-এর মাধ্যমে.
- Convection: skin/clothing-এর সাথে contact air; air movement বাড়লে convective exchange বাড়ে.
- Radiation: contact ছাড়া body ↔ surrounding surface/sun.

- Evaporation: sweat, perspiration and exhalation; dry air evaporation বাড়ায়.

### Repeated theory

Heat loss in different climatic situations repeatedly appears. Core logic: convection stops when air temperature approaches skin temperature; radiation depends on surrounding surface temperature; hot/humid conditions make evaporation + air movement critical.

### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                                                             |
|------|----------------------------------------------------------------------------------------------|
| 2016 | Explain human body heat exchange process.                                                    |
| 2018 | Describe heat loss in climatic situations; explain human heat exchange with sketches.        |
| 2020 | Human body heat exchange with environment; relation between heat production and dissipation. |
| 2021 | Thermal balance equation of human body.                                                      |
| 2024 | Vasomotor adjustment and human thermal regulation.                                           |

## A5. Ventilation, Air-space & Infiltration

**Air space per person = (Volume of built space – Volume of people & furniture) / Number of users**

Unit: m<sup>3</sup>/person.

*Directly stated in uploaded source*

**Q<sub>total</sub> = N × q<sub>person</sub>**

Total fresh-air supply = users × required fresh-air rate per person.

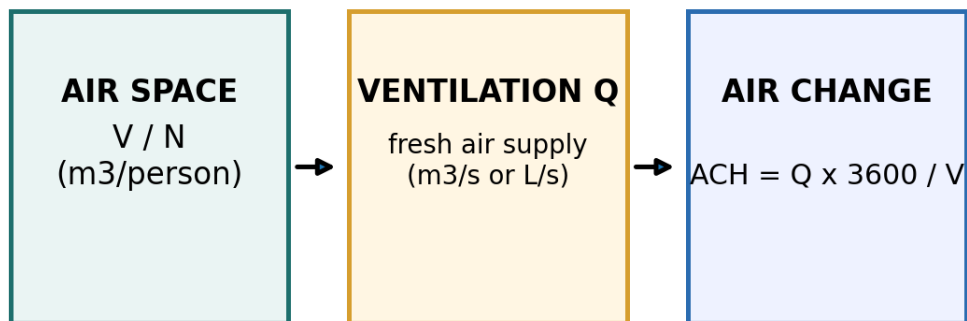
*Derived for solving repeated QB numericals*

**ACH = (Q × 3600) / V**

Q in m<sup>3</sup>/s; V = room volume in m<sup>3</sup>; result = air changes/hour.

*Derived for solving repeated QB numericals*

### Ventilation calculation chain



Ventilation calculation chain used by repeated final-term numericals.

### Recommended fresh-air table from lecture

| Air-space/person (m <sup>3</sup> ) | Minimum (L/s) | Smoking zone (L/s) | Non-smoking (L/s) |
|------------------------------------|---------------|--------------------|-------------------|
| 3                                  | 11.3          | 22.6               | 17.0              |
| 6                                  | 7.1           | 14.2               | 10.7              |
| 9                                  | 5.2           | 10.4               | 7.8               |
| 12                                 | 4.0           | 8.0                | 6.0               |

### Question Bank Match

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| Year | Focus / question                                                                                    |
|------|-----------------------------------------------------------------------------------------------------|
| 2021 | Room ventilation in m <sup>3</sup> /s, L/s and air changes/hour; fire shelter; lab/stall occupancy. |
| 2022 | Room fresh-air numerical; air-space and infiltration heat loss.                                     |
| 2023 | Room occupancy/infiltration, fire shelter and ventilation.                                          |
| 2024 | Lab ventilation and fire shelter; air-space/infiltration heat-loss question.                        |

## A6. Heat Loss due to Air Change

**Q<sub>air</sub> = (ρ × V × n × c<sub>p</sub> × ΔT) / 3600**

V = room volume; n = air changes/hour; ρ = air density; c<sub>p</sub> = specific heat; ΔT = indoor-outdoor temperature difference.

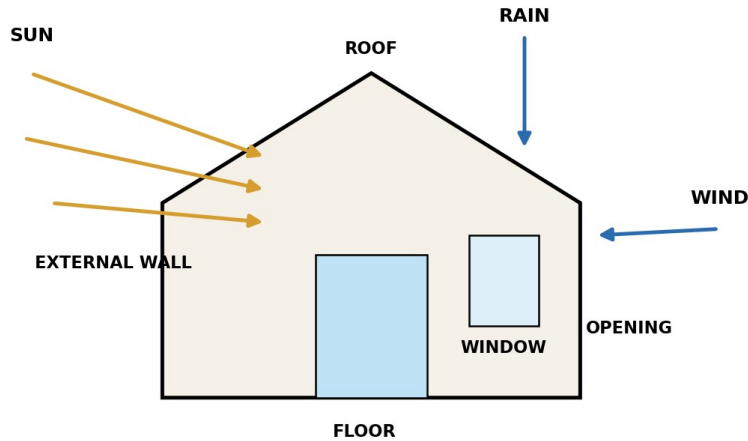
*Derived for solving repeated QB numericals*

Derivation idea: প্রতি ঘণ্টায় replaced air volume =  $nV$ . Mass/hour =  $\rho nV$ . Heat/hour = mass  $\times c_p \times \Delta T$ . প্রতি second-এ rate পেতে 3600 দিয়ে ভাগ।

#### Source note

Repeated QB numerical gives room volume, air changes/hour,  $\Delta T$  and specific heat. The uploaded slide does not print the final heat-loss equation; the above is an explicitly marked reconstruction from the heat-energy relation used to solve that question.

## A7. Microclimate & Building Envelope



**Microclimate design moderates heat, moisture, solar radiation and airflow at the envelope**

Roof, wall, floor, opening and other envelope components respond differently to heat, humidity, wind and rain.

- Roof: heat enters mainly by conduction after solar heating; time lag and heat conserving capacity matter.
- External walls: orientation changes solar exposure; east/west walls receive strong low-angle sun.
- Window/opening: admits solar heat, daylight, rain/fog and also acts as infiltration path.
- Ventilator/clerestory: higher-level hot-air exhaust can improve ventilation.
- Design tools: material, orientation, opening, architectural design, micro-planning, semi-mechanized techniques.

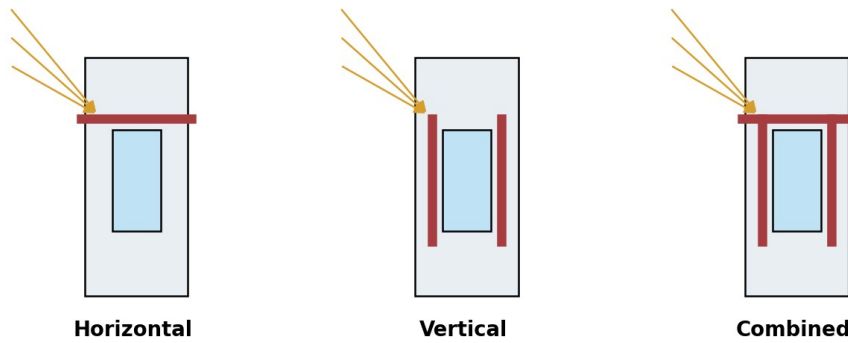
#### Question Bank Match

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| Year | Focus / question                                                             |
|------|------------------------------------------------------------------------------|
| 2018 | How climatic factors affect building envelope in designing microclimate?     |
| 2021 | Micro-climatic solution for hot-dry and warm-humid climate.                  |
| 2022 | Microclimatic solution / climate-responsive design.                          |
| 2024 | Explain climate factors affecting building envelopes in microclimate design. |

## A8. Shading Devices

### Basic shading-device types



Horizontal, vertical and combined external shading devices.

- Purpose: control solar radiation, wind flow, rain water and visibility.
- Solar heat enters windows as direct, reflected and diffuse radiation.
- External shading can intercept all three before heat enters the glazing.
- Types: horizontal, vertical, combined.

#### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                               |
|------|----------------------------------------------------------------|
| 2016 | Specify types of shading devices with sketches.                |
| 2022 | How to design shading devices for indoor comfort.              |
| 2023 | Solar shading design numerical.                                |
| 2024 | Design an overhang/shading device; solar projection numerical. |

### Projection calculation formula: not stated in uploaded Section A shading slides

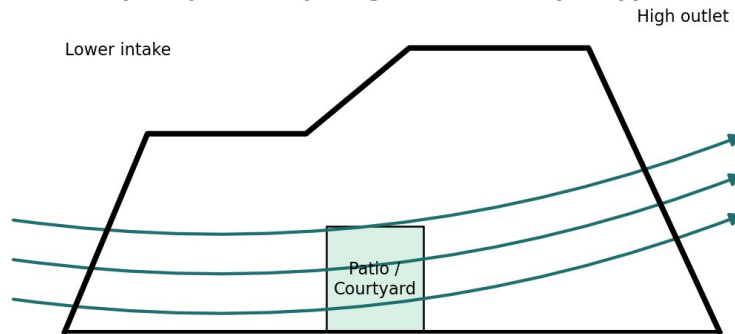
Use instructor-verified sun-angle / shadow-angle method for the 2023-2024 numerical.

*Question appears, but formula not stated in uploaded slides*

## A9. Climate-friendly Movements

### Tube House

**Tube-house concept: adjustable openings + section shape support natural ventilation**

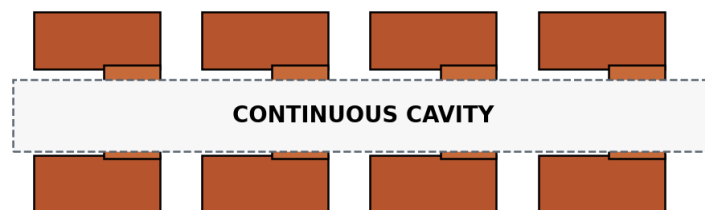


Tube-house section concept: section form and adjustable openings support natural air movement.

- Prototype housing idea; section/profile encourages natural ventilation.
- Openings can be adjusted for different seasons/time of day.
- Courtyard/patio and high-low airflow paths can support passive comfort.

### Rat-trap Bond

**Rat-trap bond - bricks on edge create a cavity while retaining wall thickness**



Exam sketch: show brick leaves, cavity, solid courses at sill/lintel/top/bottom

Brick-on-edge construction creates an internal cavity.

- Bricks placed vertically/on edge create a cavity while wall width remains about 230 mm.
- Lecture notes cite reduced material use and improved thermal/sound insulation.
- Solid courses are required at top/bottom, sill/lintel and around openings.

### Green vs Sustainable Building

- Green building focuses on measurable environmental performance: less water, energy efficiency, resource conservation, less waste, healthier spaces.
- Sustainable building additionally balances environmental, social and economic dimensions.

#### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                                         |
|------|--------------------------------------------------------------------------|
| 2016 | Tube house, green building, contemporary materials.                      |
| 2017 | Green building movement; contemporary vs traditional building materials. |
| 2021 | Tube house ventilation; rat-trap bond.                                   |
| 2023 | Rat-trap bond; sustainability parameters.                                |

| Year | Focus / question                                    |
|------|-----------------------------------------------------|
| 2024 | Tube house and sustainability of academic building. |

## PART III — CURRENT SECTION B | Billah / Safayet Sir Topics

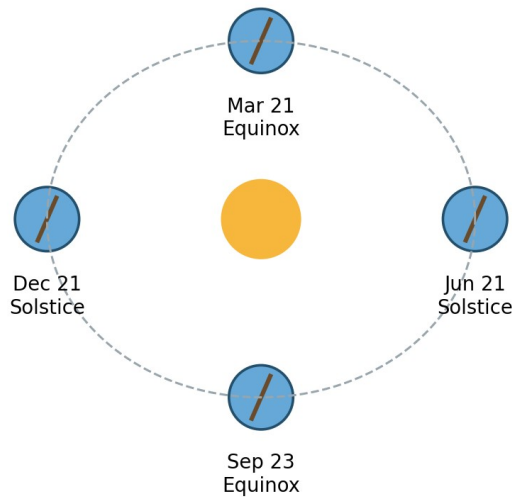
### Section B map

Climatic study • Global climatic factors • Elements/measurement • Driving rain • Tropical classification • Regional/site/urban climate • Thermal comfort and human heat balance.

### B1. Solar Radiation & Earth-Sun Relation

- Solar radiation is the dominant climatic energy source.
- Slide spectrum: about 290-2300 nm; UV 290-380 nm, visible 380-700 nm, short IR 700-2300 nm.
- Solar constant in lecture:  $1395 \text{ W/m}^2$  at top of atmosphere.
- Earth axis is  $66.5^\circ$  to orbital plane; direct-sun point migrates between Tropic of Cancer and Capricorn, producing seasons.
- Equinox around 21 March and 23 September: solar radiation normal to equator.

#### Earth-Sun Relationship: tilt controls seasonal solar distribution



Earth-axis tilt explains seasonal migration of direct solar radiation.

#### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                                       |
|------|------------------------------------------------------------------------|
| 2017 | Evaluate Earth-axis relationship from viewpoint of climatic variation. |
| 2022 | Earth-Sun relationship and global climatic variations.                 |
| 2024 | How earth-axis tilt affects solar distribution and causes seasons.     |

### B2. Radiation at Earth Surface — Cosine + Atmospheric Depletion

$$I\beta = I_n \cos \beta$$

Same cosine relation appears in Global Climatic Factors lecture.

*Directly stated in uploaded source*

- Atmospheric depletion: ozone, vapour and dust absorb/scatter part of radiation.
- Duration of sunshine controls how long radiation can be received.
- Surface angle controls received intensity through cosine law.

#### Question Bank Match

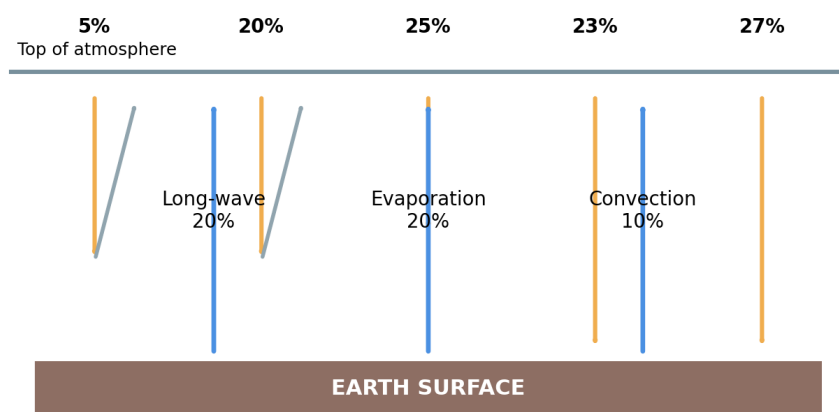
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form-এ normalised করা।

| Year | Focus / question                                             |
|------|--------------------------------------------------------------|
| 2017 | Cosine law and atmospheric depletion.                        |
| 2023 | Atmospheric depletion short note.                            |
| 2024 | Relation of earth-surface radiation to global-scale climate. |

## B3. Earth's Thermal Balance

**Incoming solar radiation: 50% reaches ground (23% diffuse + 27% direct)**



**Outgoing total from ground = 50%**

Course-slide accounting of incoming and outgoing radiation/heat flows.

| Incoming solar energy   | %  |
|-------------------------|----|
| Reflected from ground   | 5  |
| Reflected from cloud    | 20 |
| Absorbed in atmosphere  | 25 |
| Diffuse reaching ground | 23 |
| Direct reaching ground  | 27 |

| Outgoing from ground       | %  |
|----------------------------|----|
| Long-wave radiation        | 20 |
| Evaporation then radiation | 20 |
| Convection then radiation  | 10 |

### Balance logic

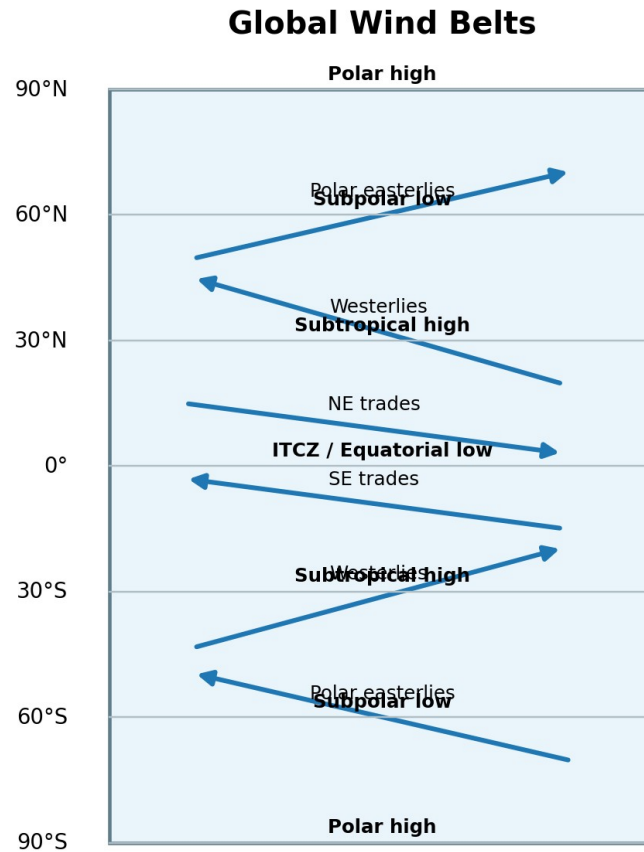
Ground receives  $23\% + 27\% = 50\%$ . Ground sends back  $20\% + 20\% + 10\% = 50\%$ . তাই দীর্ঘমেয়াদে ground heat budget balanced দেখানো হয়েছে।

### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                                             |
|------|------------------------------------------------------------------------------|
| 2016 | How earth keeps thermal balance by equating incoming and outgoing radiation. |
| 2020 | How earth keeps heat balance by equating incoming and outgoing radiation.    |
| 2021 | Earth heat balance equation / incoming-outgoing radiation.                   |

## B4. Global Wind Pattern, Thermal Force, Coriolis & ITCZ

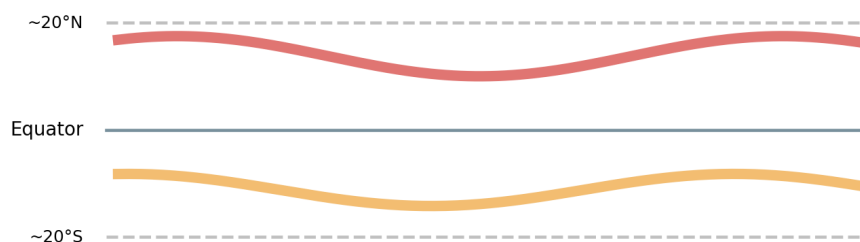


Thermal force creates pressure belts; Coriolis force deflects the flow

Global pressure/wind belts and circulation cells.

- Equator gets intense heating ~~warm~~ air rises ~~equatorial~~ low pressure.
- Subtropical region: cooled upper air sinks ~~high~~ pressure.
- Thermal force drives convection from pressure/temperature differences.
- Earth rotation modifies motion; Coriolis deflects winds. Trade winds are the resultant of thermal and Coriolis effects.
- At equator, N and S trade winds converge: Inter-Tropical Convergence Zone (ITCZ), often calm/light irregular breeze = doldrums.

**Annual ITCZ Shift follows the zone of maximum solar heating**



ITCZ follows maximum solar heating north-south seasonally.

### Shortcut - 0-30-60-90

0° Low, 30° High, 60° Low, 90° High - pressure belts মনে রাখার দ্রুত sequence.

### Question Bank Match

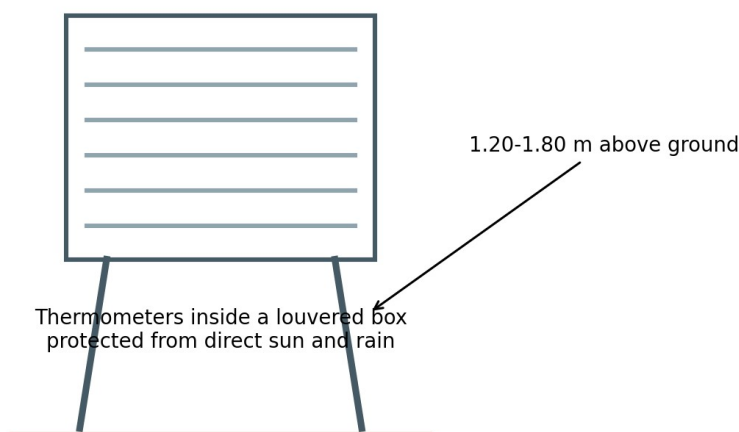
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| Year | Focus / question                                                            |
|------|-----------------------------------------------------------------------------|
| 2016 | Forces interacting to generate global wind pattern; north/south hemisphere. |
| 2021 | ITCZ and annual wind shifts; global wind pattern at different latitudes.    |
| 2022 | Thermal force + Coriolis force; westerlies/polar winds.                     |
| 2023 | Doldrums; Coriolis and wind pattern.                                        |
| 2024 | Global wind patterns at different latitudes in both hemispheres.            |

## B5. Elements of Climate & Measurement

| Element         | Instrument / expression                                   | Exam note                                                     |
|-----------------|-----------------------------------------------------------|---------------------------------------------------------------|
| Temperature     | Thermometer / thermograph; °C                             | True air temperature in Stevenson screen, approx. 1.20-1.80 m |
| Humidity        | Wet & dry-bulb hygrometer / psychrometric chart; %RH      | AH, RH, vapour pressure                                       |
| Precipitation   | Rain gauge; mm/time                                       | 24-h max for flood; hourly max for drainage                   |
| Solar radiation | Sunshine recorder for duration; radiometers for intensity | Duration $\neq$ intensity                                     |
| Wind            | Anemometer for speed; wind vane for direction; wind rose  | Open flat country usually ~10 m measurement height            |
| Sky condition   | Cloud cover fractions / percentage                        | Often two observations/day                                    |
| Vegetation      | Observation                                               | Affects shading, humidity, wind and site response             |

### STEVENSON SCREEN



Stevenson screen protects temperature instruments from direct solar/rain while allowing airflow.

$$P = P_a + P_v$$

Total atmospheric pressure = dry-air partial pressure + water-vapour partial pressure.

*Directly stated in uploaded source*

$$RH = (AH / SH) \times 100 \%$$

Relative humidity relation.

*Directly stated in uploaded source*

### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                          |
|------|-----------------------------------------------------------|
| 2018 | Stevenson screen; climatic measurement.                   |
| 2021 | Stevenson screen; driving rain; solar radiation quantity. |

| Year | Focus / question                                                             |
|------|------------------------------------------------------------------------------|
| 2022 | Instrument to measure open-field temperature; Stevenson screen.              |
| 2024 | How climatic parameters - temperature, humidity and air flow - are measured. |

## B6. Driving Rain Index (DRI)

$$\text{DRI} = \text{Annual Rainfall (m)} \times \text{Annual Average Wind Speed (m/s)}$$

Unit = m<sup>2</sup>/s.

*Directly stated in uploaded source*

| DRI | Exposure  | Design implication                               |
|-----|-----------|--------------------------------------------------|
| ≤ 3 | Sheltered | lower combined wind-rain exposure                |
| 3-7 | Moderate  | careful detailing of joints/sills/reveals        |
| > 7 | Severe    | highly exposed; strong weatherproofing/detailing |

### Shortcut - 3 and 7

DRI boundary শুধু 3 ও 7 মনে রাখো: 3 sheltered, 3-7 moderate, >7 severe.

### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                               |
|------|----------------------------------------------------------------|
| 2021 | Driving rain and protective measures against overcooling/rain. |
| 2022 | Define driving rain and design measures.                       |
| 2023 | Driving rain: definition and design exposure.                  |
| 2024 | Define driving rain; design building exposure against it.      |

## B7. Tropical Climate Classification

| Main group          | Sub-group / related type             | Core signature                                             |
|---------------------|--------------------------------------|------------------------------------------------------------|
| Warm-humid          | Equatorial; island/trade-wind        | High RH, high rain, small diurnal range, cloudy            |
| Hot-dry             | Desert/semi-desert; maritime desert  | Low RH, low rain, strong direct solar, large diurnal range |
| Composite / Monsoon | May include tropical upland subgroup | Seasonal switch: hot-dry ↔ warm-humid; sometimes cool-dry  |

### Warm-humid vs Hot-dry — exam comparison

| Factor            | Warm-humid                       | Hot-dry                             |
|-------------------|----------------------------------|-------------------------------------|
| Mean temp / range | High but relatively narrow range | Very hot day; large day-night range |
| RH                | ~75%, often 55-100%              | ~10-55%                             |
| Rain              | ~2000-5000 mm/year               | ~50-155 mm/year in slide            |
| Sky               | Cloudy 60-90%                    | Mostly clear; dust haze possible    |
| Solar             | Scattered/diffuse important      | Strong direct radiation             |
| Wind              | Often low; squalls possible      | Hot, dusty local winds              |
| Vegetation        | Rapid growth; high water table   | Sparse; dry soil                    |

### Question Bank Match

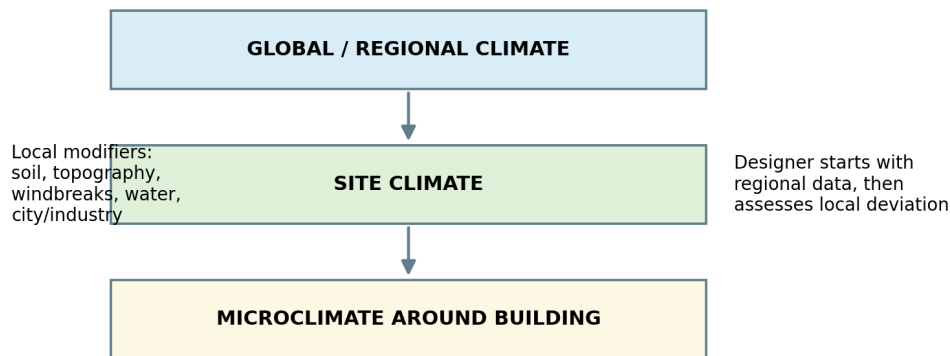
বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                         |
|------|------------------------------------------|
| 2016 | Compare Hot-dry and Warm-humid climates. |
| 2017 | Characterize composite/monsoon climate.  |
| 2018 | Characteristics of tropical climates.    |

| Year | Focus / question                                                      |
|------|-----------------------------------------------------------------------|
| 2021 | Atkinson classification; climatic design in extreme humidity.         |
| 2022 | Classification of tropical climate; warm-humid and hot-dry features.  |
| 2023 | Formation/classification of tropical climate; warm-humid and hot-dry. |
| 2024 | Warm-humid region and large day-night range in desert region.         |

## B8. Regional, Site & Urban Climate

### Climate scale and deviation



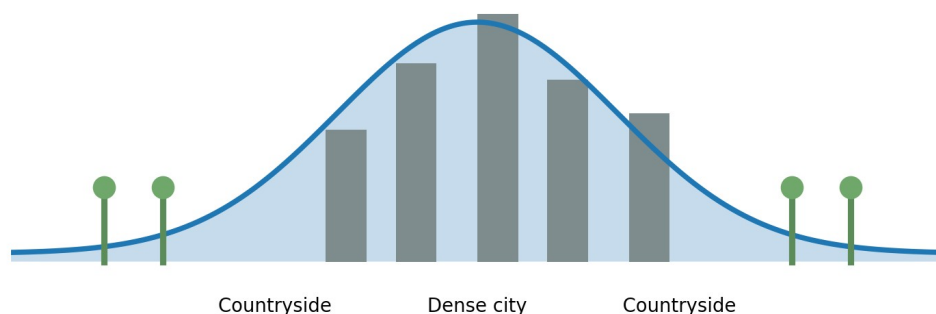
Regional climate is modified by progressively more local factors.

| Factor level | Examples                                                                                  |
|--------------|-------------------------------------------------------------------------------------------|
| Primary      | Global climatic factors; temperature, humidity, airflow as core climatic variables        |
| Secondary    | Location, diurnal variation, altitude                                                     |
| Tertiary     | Soil, topography, natural/man-made windbreakers, water bodies, cities/industries, deserts |
| Local/minor  | Ocean current, sky condition, smog, smoke, fog, thunder/lightning, sand storm             |

### Urban climate

- Changed surface qualities increase absorption and reduce evaporation.
- Buildings cast shadows, block/channel wind and store heat.
- Energy seepage from AC, appliances, engines and industry adds heat.
- Pollution can reduce outgoing radiation.
- Lecture slide notes city-country temperature deviation up to about 8°C, RH 5-10% lower and wind may reduce to less than half.

### Urban climate: heat storage + less evaporation + anthropogenic heat



Urban intervention modifies temperature, humidity, radiation and wind.

### Question Bank Match

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                                             |
|------|------------------------------------------------------------------------------|
| 2016 | Local factors responsible for site/microclimate; architect responsibilities. |
| 2017 | Microclimate; site climate; ocean current/natural wind breaker.              |
| 2018 | Site climate and temperature inversion.                                      |
| 2020 | Site design process and regional factors.                                    |
| 2021 | Urban vs macroclimate; site factors.                                         |
| 2022 | Site vs regional climate; urban climate deviation.                           |
| 2023 | Residential site design; urban deviation.                                    |
| 2024 | Site climate/local deviation; urban climate depends on human intervention.   |

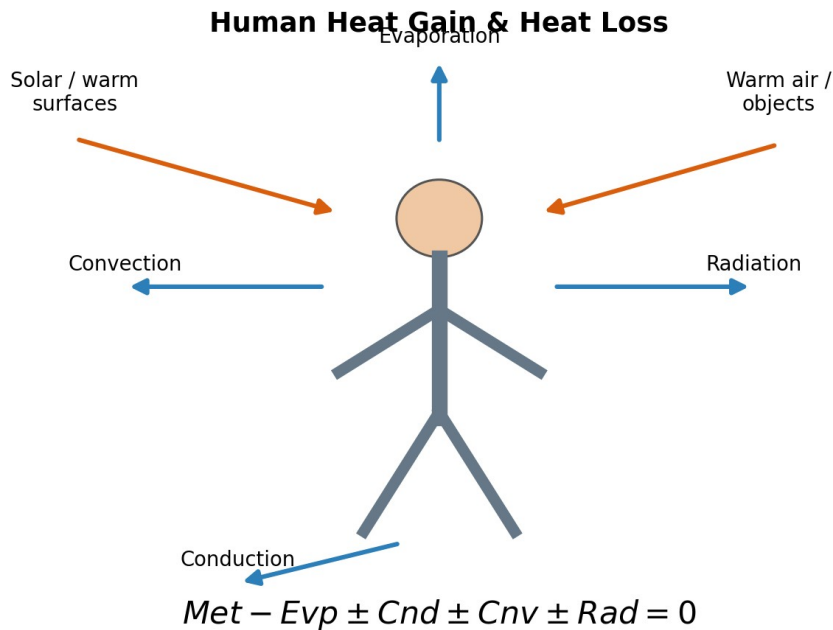
## B9. Thermal Comfort & Human Heat Balance

- Metabolism converts food into living matter and energy; basal and muscular metabolism are two divisions.
- Lecture slide: about 20% of produced heat is used internally; remaining 80% is surplus and must be dissipated.
- Heat gain modes: metabolism, conduction from warm contact object, convection from warm air, radiation from hot objects/sun.
- Heat loss: convection, radiation, evaporation and lesser conduction.
- Deep-body temperature stays around 37°C; steady state requires surplus heat to be dissipated.

$$\text{Met} - \text{Evp} \pm \text{Cnd} \pm \text{Cnv} \pm \text{Rad} = 0$$

Steady-state body heat balance.

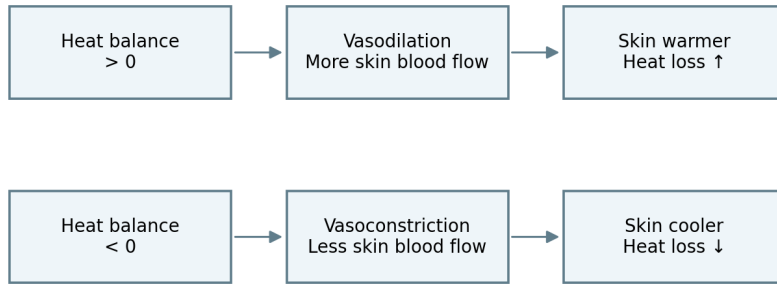
*Directly stated in uploaded source*



Human heat-balance equation groups metabolic heat with environmental exchange terms.

### Objective vs Subjective variables

| Objective / climatic | Subjective / individual           |
|----------------------|-----------------------------------|
| Radiation            | Clothing                          |
| Air temperature      | Acclimatization                   |
| Humidity             | Age and sex                       |
| Air movement         | Body shape / subcutaneous fat     |
|                      | Health, food & drink, skin colour |

**Thermoregulatory response (exam flowchart)**

If vasomotor control is insufficient: overheating → sweating; underheating → shivering

Vasomotor regulation changes skin blood flow before sweating/shivering mechanisms intensify.

**Question Bank Match**

বছর অনুযায়ী একই concept-এর repeated/closely related wording দেখানো হয়েছে। Exact wording scan থেকে readable form-এ normalised করা।

| Year | Focus / question                                                                |
|------|---------------------------------------------------------------------------------|
| 2018 | Thermal comfort; body heat production/release; subjective variables; ET.        |
| 2020 | Human heat exchange and heat balance.                                           |
| 2021 | Human thermal balance equation.                                                 |
| 2022 | Human heat production/release; subjective variables.                            |
| 2023 | Objective variables; thermal comfort; heat-balance regulatory mechanism.        |
| 2024 | How body maintains thermal balance; subjective variables; vasomotor adjustment. |

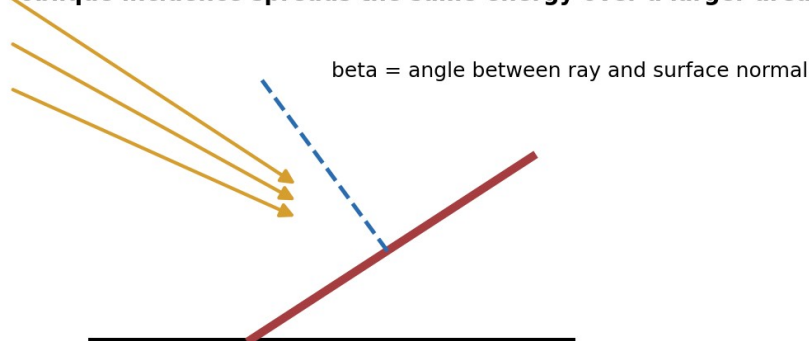
## PART IV — PROOF / DERIVATION BANK

### How to write derivation in exam

প্রথম line-এ law/definition লিখবে, প্রতিটি symbol define করবে, diagram থাকলে draw করবে, algebra 3-5 steps-এ রাখবে, শেষে boxed final relation + physical meaning লিখবে।

### D1. Cosine Law — derivation

**Cosine Law - oblique incidence spreads the same energy over a larger area**



$$I_{\text{tilted}} = I_{\text{normal}} \times \cos(\beta)$$

5. ধরা যাক parallel solar beam-এর normal intensity =  $I_n$  এবং surface normal-এর সাথে incidence angle =  $\beta$ .
6. একই beam bundle normal surface-এ projected area B এবং tilted surface-এ larger actual area C-তে ছড়িয়ে পড়ে.
7. Geometry থেকে  $\cos \beta = B/C$ , অর্থাৎ  $B = C \cos \beta$ .
8. একই total radiation energy হলে intensity  $\propto 1/\text{area}$ . তাই tilted surface-এর intensity =  $I_n \times (B/C)$ .
9.  $B/C = \cos \beta$  বসালে  $I_\beta = I_n \cos \beta$ .

$$I_\beta = I_n \cos \beta$$

$\beta = 0^\circ \Rightarrow$  maximum intensity;  $\beta$  increases  $\Rightarrow$  intensity falls.

*Directly stated in uploaded source*

### D2. Relative Humidity relation — definition proof

10. AH = given temperature-এ air-এ actual moisture content.
11. SH = same temperature-এ maximum/saturation moisture capacity.
12. Relative humidity মানে actual capacity-এর fraction, expressed as percentage.
13. Fraction =  $AH/SH$ . Percentage করতে  $\times 100$ .

$$RH = (AH / SH) \times 100 \%$$

All humidity quantities must be expressed on the same mass/volume basis.

*Directly stated in uploaded source*

### D3. Earth's Thermal Balance — accounting proof

14. Incoming solar energy = 100% ধরা হয়.
15. 5% ground reflection + 20% cloud reflection + 25% atmosphere absorption = 50% diverted before/around ground heating.
16. Ground receives diffuse 23% + direct 27% = 50%.
17. Ground loses 20% long-wave + 20% evaporation then radiation + 10% convection then radiation = 50%.
18. Therefore ground input = ground output = 50%; long-term thermal balance is maintained in the lecture model.

$$\text{Ground gain} = 23\% + 27\% = 50\% = 20\% + 20\% + 10\% = \text{Ground loss}$$

## D4. Human Heat Balance — energy-conservation derivation

19. Metabolism (Met) produces internal body heat.
20. Evaporation (Evp) always removes latent heat, so it is written as a loss term.
21. Conduction (Cnd), convection (Cnv) and radiation (Rad) may be gain or loss depending on environmental temperature gradients; hence  $\pm$  signs.
22. Steady deep-body temperature means net heat storage = 0.
23. Therefore all gains and losses algebraically sum to zero.

$$\text{Met} - \text{Evp} \pm \text{Cnd} \pm \text{Cnv} \pm \text{Rad} = 0$$

If the sum departs from zero, vasomotor adjustment occurs; persistent overheating leads to sweating, underheating to shivering.

Directly stated in uploaded source

## D5. Air Changes per Hour (ACH) — derivation

24.  $Q$  = ventilation flow rate in  $\text{m}^3/\text{s}$ .
25. One hour = 3600 s,  $\therefore$  one hour- $\Delta$  supplied/replaced air volume =  $3600Q \text{ m}^3$ .
26. Room volume =  $V \text{ m}^3$ .
27. Number of complete room-volume replacements per hour = supplied volume per hour / room volume.
28. Hence  $\text{ACH} = 3600Q / V$ .

$$\text{ACH} = 3600 Q / V$$

$Q$  in  $\text{m}^3/\text{s}$ ,  $V$  in  $\text{m}^3$ .

Derived for solving repeated QB numericals

## D6. Total Fresh-air Requirement — derivation

29. If each person needs  $q_{\text{person}} \text{ L/s}$ ,  $N$  persons need  $N$  identical shares.
30. Total required supply = sum of individual supplies =  $N \times q_{\text{person}}$ .
31. Convert  $\text{L/s}$  to  $\text{m}^3/\text{s}$  by dividing by 1000.

$$Q_{\text{total}} = N q_{\text{person}} ; \quad Q(\text{m}^3/\text{s}) = Q(\text{L/s}) / 1000$$

Use the lecture fresh-air table to choose  $q_{\text{person}}$ .

Derived for solving repeated QB numericals

## D7. Heat Loss due to Air Change — derivation

32. Room volume  $V$ , air changes  $n$  per hour  $\Rightarrow$  air volume replaced per hour =  $nV$ .
33. If air density =  $\rho$ , mass replaced per hour =  $\rho nV$ .
34. For temperature difference  $\Delta T$ , heat carried per hour = mass  $\times c_p \times \Delta T = \rho nV c_p \Delta T$ .
35. Heat-loss rate per second = divide by 3600.
36. Thus  $\dot{Q}_{\text{air}} = \rho V n c_p \Delta T / 3600$ .

$$\dot{Q}_{\text{air}} = \rho V n c_p \Delta T / 3600$$

Standard reconstruction for the repeated air-change heat-loss numerical; not printed verbatim in uploaded slide.

Derived for solving repeated QB numericals

## D8. Driving Rain Index — relation, not a mathematical proof

$$\text{DRI} = R \times V_w$$

$R$  = annual rainfall (m),  $V_w$  = annual average wind speed (m/s).

Directly stated in uploaded source

এটি lecture-এ location exposure index হিসেবে দেওয়া empirical relation. তাই “proof” করার বদলে physical interpretation লিখবে: বেশি rain □ বেশি water load; বেশি wind rain আরও oblique/forcefulভাবে envelope-এ আঘাত করে; দুটির product combined exposure বোঝায়।

## PART V — MOST REPEATED THEORY QUESTION BANK

### Priority rule

নিচের list exact marks prediction নয়। Uploaded 2016-2024 papers-এ recurring concept অনুযায়ী High / Very High priority করা হয়েছে। একই concept বিভিন্ন বছরে wording বদলে এসেছে।

### Current Section A — prioritized theory list

| Priority  | Topic / normalized question                                                         | Seen in papers                                 |
|-----------|-------------------------------------------------------------------------------------|------------------------------------------------|
| VERY HIGH | Human thermal comfort; body heat gain/loss; heat-balance / vasomotor regulation     | 2016, 2017, 2018, 2020, 2021, 2022, 2023, 2024 |
| VERY HIGH | Ventilation, air-space, infiltration, fresh-air requirements; associated numericals | 2021, 2022, 2023, 2024                         |
| HIGH      | Microclimate design: envelope, climatic factors, design tools/solutions             | 2018, 2021, 2022, 2024                         |
| HIGH      | Shading devices: types, external shading, design significance                       | 2016, 2022, 2023, 2024                         |
| HIGH      | ET/CET/EW/OT, cosine law, atmospheric depletion                                     | 2016, 2017, 2021, 2022, 2023, 2024             |
| HIGH      | Tube house, rat-trap bond, green/sustainable building                               | 2016, 2017, 2021, 2022, 2023, 2024             |
| MED-HIGH  | Climate-Context-Comfort / built environment / designer responsibility               | 2016, 2017, 2018, 2024                         |
| MED-HIGH  | Climatic measurement: Stevenson screen, humidity relation                           | 2018, 2020, 2021, 2022, 2024                   |

### Current Section B — prioritized theory list

| Priority  | Topic / normalized question                                                | Seen in papers                                 |
|-----------|----------------------------------------------------------------------------|------------------------------------------------|
| VERY HIGH | Regional/site/micro/urban climate and factors causing deviation            | 2016, 2017, 2018, 2020, 2021, 2022, 2023, 2024 |
| VERY HIGH | Tropical climate classification; warm-humid vs hot-dry vs composite        | 2016, 2017, 2018, 2021, 2022, 2023, 2024       |
| VERY HIGH | Global wind pattern, thermal force, Coriolis, ITCZ/doldrums, annual shifts | 2016, 2021, 2022, 2023, 2024                   |
| HIGH      | Thermal comfort, body heat balance, objective/subjective variables         | 2018, 2020, 2021, 2022, 2023, 2024             |
| HIGH      | Elements of climate and measuring climate; Stevenson, wind, precipitation  | 2018, 2021, 2022, 2024                         |
| HIGH      | Driving rain and envelope exposure design                                  | 2021, 2022, 2023, 2024                         |
| MED-HIGH  | Earth axis, solar radiation quality/quantity, Earth-Sun relation           | 2017, 2021, 2022, 2024                         |
| MED-HIGH  | Earth thermal balance / incoming-outgoing radiation                        | 2016, 2020, 2021                               |

# Theory answer skeletons — 10/15/20 marks

## 10 marks

- Definition: 2-3 lines
- 4-6 key points / classification
- 1 small labelled sketch/table
- 1 design significance / example
- 1-line conclusion

## 15 marks

- Definition + context
- Mechanism/process in 6-8 points
- Comparison table / formula if relevant
- Clear sketch/diagram
- Design implication + short conclusion

## 20 marks

- Intro + precise definitions
- Detailed mechanism/factors/sub-classification
- At least one clean diagram and one table where useful
- Numerical relation / derivation if relevant
- Built-form application, limitations and conclusion

### Shortcut - D-M-S-A-C

Definition Mechanism Sketch Application Conclusion . প্রায় সব theory answer এই ৫-step skeleton-এ লেখা যায়।

## Must-draw sketches

| Sketch                        | Use in answer                            |
|-------------------------------|------------------------------------------|
| Cosine law                    | Solar radiation / incidence              |
| Earth thermal balance         | Incoming-outgoing radiation              |
| Global wind belts / cells     | Thermal force, Coriolis, ITCZ            |
| ITCZ seasonal shift           | Annual wind/rain shift                   |
| Stevenson screen              | Temperature measurement                  |
| Wind rose                     | Monthly/annual wind graph interpretation |
| Human heat exchange / balance | Thermal comfort                          |
| Vasomotor flow                | Body thermoregulation                    |
| Envelope/microclimate         | Roof-wall-window design                  |
| Shading types                 | Horizontal/vertical/combined             |
| Tube house section            | Passive ventilation                      |
| Rat-trap bond cavity          | Climate-friendly construction            |

# PART VI — LAST-MINUTE REVISION

## One-page formula memory

$$RH = AH/SH \times 100\% \quad | \quad P = P_a + P_v$$

Humidity + vapour pressure.

*Directly stated in uploaded source*

$$I\beta = I_n \cos \beta \quad | \quad DRI = R \times V_w$$

Solar incidence + wind-driven rain.

*Directly stated in uploaded source*

$$Met - Evp \pm Cnd \pm Cnv \pm Rad = 0$$

Human thermal balance.

*Directly stated in uploaded source*

$$\text{Air-space/person} = (V_{\text{space}} - V_{\text{occupants/furniture}})/N$$

Available fresh-air volume per user.

*Directly stated in uploaded source*

$$Q_{\text{total}} = Nq \quad | \quad ACH = 3600Q/V$$

Ventilation chain.

*Derived for solving repeated QB numericals*

$$\dot{Q}_{\text{air}} = pVncp\Delta T/3600$$

Heat loss due to air changes.

*Derived for solving repeated QB numericals*

## One-page theory memory

| Trigger word    | Recall chain                                                                   |
|-----------------|--------------------------------------------------------------------------------|
| ET / CET        | ET = T+H+Air movement → CET adds radiation                                     |
| Warm-humid      | High RH + high rain + small range + cloudy + ventilation/shade                 |
| Hot-dry         | Low RH + low rain + large range + direct sun + thermal mass/shade              |
| Global wind     | OL → 30H → 60L → 90H; thermal + Coriolis                                       |
| ITCZ            | Trade winds meet at equator → low pressure → seasonal N-S shift                |
| Site climate    | Regional data + local deviations:<br>soil/topography/water/windbreak/urban     |
| Urban climate   | Dark hard surfaces + stored heat + waste heat + pollution + reduced vegetation |
| Thermal comfort | Metabolism + C/R/C/E exchange + vasomotor/sweat/shiver                         |
| Driving rain    | Rain × wind; 3 and 7 exposure boundaries                                       |
| Microclimate    | Material + orientation + opening + architecture + micro-planning               |

### Final exam warning

Shading effective-projection numerical, U-value/time-lag/decrement-factor formulae এবং অন্য কোনো relation যা uploaded current slides-এ explicit নয় - এই booklet তা invent করেনি। Past-paper wording দেখলে instructor/class-solved method follow করবে.

End of Master Book

Made by 2223033